# Homework 2

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# Problem 1

Prove that if $A$ and $B$ are both countable sets and $A \cap B = \varnothing$, then $A \cup B$ is countable. (Hint: We proved that $\mathbb{Z}$ is countable in class. Try using a similar idea here.)

**Discussion & Scratch Work**

Because $A$ and $B$ are countable, there exist enumerations

$$A = \{a_1, a_2, a_3, \dots\}, \qquad B = \{b_1, b_2, b_3, \dots\}.$$

Intuitively, this means we can list the elements of $A$ and $B$ in a sequence. To show that $A \cup B$ is countable, we want to construct a single list that contains every element of $A$ and $B$ exactly once. The idea is to interleave the two lists, alternating between an element of $A$ and an element of $B$. The assumption $A \cap B = \varnothing$ ensures that no element appears in both lists, so this alternating procedure never produces duplicates.

**Proof.**
Since $A$ and $B$ are countable, there exist bijections

$$f : \mathbb{N} \to A \quad \text{and} \quad g : \mathbb{N} \to B.$$

Write $a_n = f(n)$ and $b_n = g(n)$, so that

$$A = \{a_1, a_2, a_3, \dots\} \quad \text{and} \quad B = \{b_1, b_2, b_3, \dots\}.$$

Define a function $h : \mathbb{N} \to A \cup B$ by

$$h(2n-1) = a_n \quad \text{and} \quad h(2n) = b_n \qquad (n \in \mathbb{N}).$$

We show that $h$ is bijective.

First, $h$ is surjective. Let $x \in A \cup B$. If $x \in A$, then $x = a_n$ for some $n$, and hence $x = h(2n-1)$. If $x \in B$, then $x = b_n$ for some $n$, and hence $x = h(2n)$. Thus every element of $A \cup B$ is in the image of $h$.

Next, $h$ is injective. Suppose that $h(k) = h(\ell)$. If $k$ and $\ell$ have different parity, then one of $h(k)$ lies in $A$ and the other lies in $B$, which is impossible because $A \cap B = \varnothing$. Hence $k$ and $\ell$ must have the same parity. If $k = 2m-1$ and $\ell = 2n-1$, then $a_m = a_n$, and since $f$ is injective, we have $m = n$, so $k = \ell$. If $k = 2m$ and $\ell = 2n$, then $b_m = b_n$, and since $g$ is injective, we again obtain $m = n$, so $k = \ell$.

Therefore $h$ is a bijection from $\mathbb{N}$ onto $A \cup B$. It follows that $A \cup B$ is countable. $\square$

# Problem 2

Let $A_n$ be a countable set for each $n \in \mathbb{N}$, and suppose that $A_m \cap A_n = \varnothing$ whenever $m \neq n$. Prove that the union $A = A_1 \cup A_2 \cup A_3 \cup \cdots$ is countable. (Hint: We proved that $\mathbb{N} \times \mathbb{N}$ is countable in class. Try using a similar idea here.)

**Discussion & Scratch Work**

Because each $A_n$ is countable, we can "list" its elements as

$$A_n = \{a_{n,1}, a_{n,2}, a_{n,3}, \dots\}.$$

So every element of the big union $A = \bigcup_{n\in\mathbb{N}} A_n$ comes with two pieces of information: (1) which set $A_n$ it belongs to, and (2) its position in the list of $A_n$.

That suggests encoding an element of $A$ by a pair $(n, k) \in \mathbb{N} \times \mathbb{N}$, where $n$ tells us which $A_n$ we are in, and $k$ tells us which element of $A_n$ we chose. Since we proved in class that $\mathbb{N} \times \mathbb{N}$ is countable, we can list all pairs $(n, k)$ in a single sequence. If we map $(n, k)$ to the $k$-th element of $A_n$, we get a single list that hits every element of $A$.

The disjointness assumption $A_m \cap A_n = \varnothing$ for $m \neq n$ is what prevents duplicates from coming from different $A_n$'s.

**Proof.**
For each $n \in \mathbb{N}$, the set $A_n$ is countable, so there exists a bijection

$$f_n : \mathbb{N} \to A_n.$$

Define a function $h : \mathbb{N} \times \mathbb{N} \to A$ by

$$h(n, k) := f_n(k) \qquad ((n, k) \in \mathbb{N} \times \mathbb{N}).$$

We claim that $h$ is bijective.

*Surjectivity.* Let $x \in A$. By definition of $A = \bigcup_{n\in\mathbb{N}} A_n$, there exists some $n \in \mathbb{N}$ such that $x \in A_n$. Since $f_n$ is surjective onto $A_n$, there exists $k \in \mathbb{N}$ with $f_n(k) = x$. Thus $h(n, k) = x$, so $h$ is surjective.

*Injectivity.* Suppose $h(n, k) = h(m, \ell)$. Then

$$f_n(k) = f_m(\ell).$$

The left-hand side lies in $A_n$ and the right-hand side lies in $A_m$. If $n \neq m$, then this would imply $A_n \cap A_m \neq \varnothing$, contradicting the hypothesis that the $A_n$ are pairwise disjoint. Hence $n = m$. With $n = m$, we have $f_n(k) = f_n(\ell)$, and since $f_n$ is injective, it follows that $k = \ell$. Therefore $(n, k) = (m, \ell)$, and $h$ is injective.

So $h$ is a bijection $\mathbb{N} \times \mathbb{N} \to A$.

Finally, we proved in class that $\mathbb{N} \times \mathbb{N}$ is countable, i.e. there exists a bijection

$$\varphi : \mathbb{N} \to \mathbb{N} \times \mathbb{N}.$$

Then the composition $h \circ \varphi : \mathbb{N} \to A$ is a bijection. Therefore $A$ is countable. $\square$

# Problem 3

Recall that given two sets $A$ and $B$, we write $A \sim B$ if and only if there exists a bijective function $f : A \to B$. Prove that $\sim$ is an equivalence relation.

**Discussion & Scratch Work**

We want to prove that $\sim$ is an equivalence relation, so we must check three properties: reflexivity, symmetry, and transitivity.

The definition says $A \sim B$ means "there exists a bijection $f : A \to B$."

- **Reflexive:** any set $A$ is in bijection with itself via the identity function defined by $f : A \to A$ with $f(a) = a$ for all $a \in A$.
- **Symmetric:** if $A \sim B$, there is a bijection $f : A \to B$. A bijection has an inverse function $f^{-1} : B \to A$, and $f^{-1}$ is also a bijection, so $B \sim A$.
- **Transitive:** if $A \sim B$ and $B \sim C$, then there are bijections $f : A \to B$ and $g : B \to C$. Their composition $g \circ f : A \to C$ is again a bijection, so $A \sim C$.

So the whole proof is basically: identity gives reflexive, inverse gives symmetric, and composition gives transitive.

**Proof.**
We verify the three defining properties.

*(Reflexivity).* Let $A$ be a set. Define a function $f : A \to A$ by $f(a) = a$ for all $a \in A$. Then $f$ is bijective. Hence $A \sim A$.

*(Symmetry).* Suppose $A \sim B$. Then there exists a bijection $f : A \to B$. Since $f$ is bijective, it has an inverse function $f^{-1} : B \to A$, and $f^{-1}$ is bijective. Therefore $B \sim A$.

*(Transitivity).* Suppose $A \sim B$ and $B \sim C$. Then there exist bijections $f : A \to B$ and $g : B \to C$. The composition $g \circ f : A \to C$ is bijective, so $A \sim C$.

Thus $\sim$ is reflexive, symmetric, and transitive; hence it is an equivalence relation. $\square$

# Problem 4

For $a, b \in \mathbb{Z}$, define $a \sim b$ if and only if $b - a$ is divisible by 2.

**(a)** Prove that $\sim$ is an equivalence relation on $\mathbb{Z}$.

**(b)** Find the equivalence classes.

### Discussion & Scratch Work

**Big picture.** The relation is saying that two integers are equivalent when they have the *same parity* (both even or both odd). Another way to say this is that $a \sim b$ exactly when $a$ and $b$ differ by an *even* integer.

To prove that $\sim$ is an equivalence relation, we check the three properties from the notes:

- **Reflexive:** always true because $a - a = 0$ is divisible by 2.
- **Symmetric:** if $b - a$ is divisible by 2, then $a - b = -(b - a)$ is also divisible by 2.
- **Transitive:** if $b - a$ and $c - b$ are divisible by 2, then adding them gives $c - a = (c - b) + (b - a)$, also divisible by 2.

For part (b), once you realize this is "same parity," there should be exactly two equivalence classes: the evens and the odds. Concretely:

$$[0] = \{\text{even integers}\} = \{2k : k \in \mathbb{Z}\} = 2\mathbb{Z}, \qquad [1] = \{\text{odd integers}\} = \{2k{+}1 : k \in \mathbb{Z}\} = 2\mathbb{Z}{+}1.$$

Any even integer is equivalent to 0, and any odd integer is equivalent to 1.

**Proof.**
*(a)* We show that $\sim$ is reflexive, symmetric, and transitive on $\mathbb{Z}$.

*Reflexive.* Let $a \in \mathbb{Z}$. Then $a - a = 0$, and 0 is divisible by 2. Hence $a \sim a$.

*Symmetric.* Suppose $a \sim b$. By definition, $b - a$ is divisible by 2, so $b - a = 2k$ for some $k \in \mathbb{Z}$. Then

$$a - b = -(b - a) = -2k = 2(-k),$$

so $a - b$ is divisible by 2. Thus $b \sim a$.

*Transitive.* Suppose $a \sim b$ and $b \sim c$. Then $b - a = 2k$ and $c - b = 2\ell$ for some $k, \ell \in \mathbb{Z}$. Adding,

$$c - a = (c - b) + (b - a) = 2\ell + 2k = 2(k + \ell),$$

which is divisible by 2. Hence $a \sim c$.

Therefore $\sim$ is an equivalence relation on $\mathbb{Z}$.

*(b)* The equivalence classes are exactly the parity classes. First, if $n \in \mathbb{Z}$ is even, then $n - 0 = n$ is divisible by 2, so $n \sim 0$; hence every even integer lies in $[0]$. Conversely, if $n \in [0]$, then $n \sim 0$, so $n - 0 = n$ is divisible by 2, which means $n$ is even. Thus

$$[0] = \{2k : k \in \mathbb{Z}\} = 2\mathbb{Z}.$$

Similarly, if $n \in \mathbb{Z}$ is odd then $n-1$ is even (hence divisible by 2), so $n \sim 1$ and $n \in [1]$; and if $n \in [1]$, then $n \sim 1$ so $n-1$ is divisible by 2, meaning $n$ is odd. Hence

$$[1] = \{2k+1 : k \in \mathbb{Z}\} = 2\mathbb{Z}+1.$$

These are the two equivalence classes. $\square$

# Problem 5

Let $(F, +, \cdot)$ be a field.

**(a)** Prove that for any $x \in F$, the function $f : F \to F$, $f(y) = x + y$ is bijective.

**(b)** Prove that for any $x \in F \setminus \{0\}$, the function $g : F \to F$, $g(y) = x \cdot y$ is bijective.

### Discussion & Scratch Work

**Big picture.** In a field, "adding a fixed element" and "multiplying by a fixed nonzero element" are reversible operations.

- For (a), if we define $f(y) = x + y$, then we can undo it by adding the additive inverse $-x$:
$$f(y) = x + y \quad \Longrightarrow \quad y = (-x) + f(y).$$
So the inverse candidate is $F \to F$, $y \mapsto (-x) + y$.
- For (b), if $x \neq 0$ and we define $g(y) = x \cdot y$, then we can undo it by multiplying by $x^{-1}$:
$$g(y) = x \cdot y \quad \Longrightarrow \quad y = x^{-1} \cdot g(y).$$
So the inverse candidate is $F \to F$, $y \mapsto x^{-1} \cdot y$.

Once we exhibit an inverse function (and verify the two composition identities), bijectivity follows.

**Proof.**
*(a)* Fix $x \in F$ and define $f : F \to F$ by $f(y) = x + y$. Define $h : F \to F$ by
$$h(y) = (-x) + y.$$
We show that $h$ is the inverse of $f$ by checking both compositions.
For any $y \in F$,
$$(h \circ f)(y) = h(f(y)) = h(x + y) = (-x) + (x + y) = \big((-x) + x\big) + y = 0 + y = y,$$
using associativity of $+$, the definition of $-x$, and the additive identity.
For any $y \in F$,
$$(f \circ h)(y) = f(h(y)) = f((-x) + y) = x + ((-x) + y) = (x + (-x)) + y = 0 + y = y,$$
again by associativity and the definition of additive inverse.
Thus $h = f^{-1}$, so $f$ is bijective.

*(b)* Fix $x \in F \setminus \{0\}$ and define $g : F \to F$ by $g(y) = x \cdot y$. Since $x \neq 0$, there exists $x^{-1} \in F$ such that $x \cdot x^{-1} = x^{-1} \cdot x = 1$. Define $k : F \to F$ by
$$k(y) = x^{-1} \cdot y.$$

We again check both compositions. For any $y \in F$,

$$(k \circ g)(y) = k(g(y)) = k(x \cdot y) = x^{-1} \cdot (x \cdot y) = (x^{-1} \cdot x) \cdot y = 1 \cdot y = y,$$

using associativity of $\cdot$ and the multiplicative identity.
For any $y \in F$,

$$(g \circ k)(y) = g(k(y)) = g(x^{-1} \cdot y) = x \cdot (x^{-1} \cdot y) = (x \cdot x^{-1}) \cdot y = 1 \cdot y = y.$$

Thus $k = g^{-1}$, so $g$ is bijective. $\square$

# Problem 6

Let $(F, +, \cdot)$ be a field. Suppose that $F = \{0, 1, a, b\}$ is a set with exactly four elements, where 0 and 1 denote the identities for $+$ and $\cdot$ respectively, and $a, b$ denote the remaining two elements of $F$. Fill in the addition and multiplication tables below, and justify your answer.

| $+$ | $0$ | $1$ | $a$ | $b$ |
|---|---|---|---|---|
| $0$ | | | | |
| $1$ | | | | |
| $a$ | | | | |
| $b$ | | | | |

| $\cdot$ | $0$ | $1$ | $a$ | $b$ |
|---|---|---|---|---|
| $0$ | | | | |
| $1$ | | | | |
| $a$ | | | | |
| $b$ | | | | |

(Hint: Using the previous problem, show that each row and column in the addition table contains every element of $F$ exactly once, like a Sudoku puzzle. Show that the same is true for the nonzero rows and columns of the multiplication table. Note that for each entry, there is only one correct answer.)

**Discussion & Scratch Work**

**Strategy.** We know $F = \{0, 1, a, b\}$ is a field, so $(F, +)$ is an abelian group of order 4 and $F \setminus \{0\} = \{1, a, b\}$ is an abelian group under $\cdot$ of order 3.

**Addition table.** Since $|F| = 4 = 2^2$, the characteristic of $F$ must be 2 (the characteristic of a field is a prime $p$ and $|F|$ is a power of $p$). Hence

$$1 + 1 = 0.$$

In fact, in an additive group of order 4 with characteristic 2, every element satisfies $x + x = 0$. So

$$1 + 1 = a + a = b + b = 0.$$

Now use the "Sudoku" constraint: for each fixed $x \in F$, the map $y \mapsto x + y$ is a bijection (Problem 5 style), so each row/column of the addition table contains each element of $F$ exactly once. For example, in the row for 1, the entries are $1 + 0 = 1$ and $1 + 1 = 0$, so the remaining two entries must be $a$ and $b$. Also $1 + a$ cannot be 0 (else $a = -1 = 1$) and cannot be 1 (else $a = 0$) and cannot be $a$ (else $1 = 0$), so necessarily $1 + a = b$. Similarly $1 + b = a$, and then $a + b = 1$.

**Multiplication table.** The nonzero elements form a group of order 3, so it is cyclic. In particular, for any $x \in \{a, b\}$, we have $x^3 = 1$ and $x \neq 1$. Thus if we pick $a \neq 1$, then $a^2 \neq 1$ and must equal the remaining nonzero element $b$. Then $a^3 = a \cdot a^2 = a \cdot b = 1$, so $ab = 1$. Commutativity gives $ba = 1$. Also $b = a^2$ implies $b^2 = a^4 = a$ (since $a^3 = 1$), and the row/column "Sudoku" constraint for multiplication by a nonzero element gives that each nonzero row/column is a permutation of $\{1, a, b\}$. Finally, 0 times anything is 0.

**Proof.**
We determine the operation tables using only field axioms and the hint.

**Step 1: the addition table.** Since $F$ is a field with $|F| = 4$, its characteristic is a prime $p$ and $|F| = p^n$ for some $n \in \mathbb{N}$. Hence $4 = p^n$ forces $p = 2$. Therefore

$$1 + 1 = 0.$$

More generally, for any $x \in F$ we have $x + x = (1 + 1)x = 0 \cdot x = 0$, so

$$1 + 1 = a + a = b + b = 0.$$

Now fix any $x \in F$. By Problem 5(a) (applied in the additive group), the map $T_x : F \to F$ given by $T_x(y) = x+y$ is bijective. Hence each row and column of the addition table contains every element of $F$ exactly once.

We already know the 0-row and 0-column are identities: $0 + y = y$ and $x + 0 = x$. In the 1-row we have $1+0 = 1$ and $1+1 = 0$, so $1+a$ and $1+b$ must be the remaining two elements $a$ and $b$. But $1 + a$ cannot equal $0, 1$, or $a$ (each would force a contradiction), so $1 + a = b$. Then the remaining entry is $1 + b = a$. Finally, using commutativity, $a + b = b + a$, and using $1 + a = b$ we get

$a+b = 1$ (since adding $a$ to both sides of $1 + a = b$ gives $1 + (a + a) = b + a$, i.e. $1 = b + a$).

Thus the addition table is

| $+$ | $0$ | $1$ | $a$ | $b$ |
|---|---|---|---|---|
| $0$ | $0$ | $1$ | $a$ | $b$ |
| $1$ | $1$ | $0$ | $b$ | $a$ |
| $a$ | $a$ | $b$ | $0$ | $1$ |
| $b$ | $b$ | $a$ | $1$ | $0$ |

**Step 2: the multiplication table.** The set $F \setminus \{0\} = \{1, a, b\}$ is an abelian group under $\cdot$ of order 3, hence cyclic. So the two elements $a$ and $b$ are the two non-identity elements of this cyclic group. In particular, $a \neq 1$ and $a^3 = 1$. Since $a^2 \neq 1$ (otherwise $a$ would have order 2, impossible in a group of order 3), we must have

$$a^2 = b.$$

Then

$$a \cdot b = a \cdot a^2 = a^3 = 1,$$

so $ab = 1$. By commutativity, $ba = 1$. Also $b = a^2$ implies

$$b^2 = a^4 = a \cdot a^3 = a.$$

As usual in a field, $0 \cdot x = 0$ for all $x \in F$. Therefore the multiplication table is

| $\cdot$ | $0$ | $1$ | $a$ | $b$ |
|---|---|---|---|---|
| $0$ | $0$ | $0$ | $0$ | $0$ |
| $1$ | $0$ | $1$ | $a$ | $b$ |
| $a$ | $0$ | $a$ | $b$ | $1$ |
| $b$ | $0$ | $b$ | $1$ | $a$ |

These tables are uniquely determined by the field axioms and the "Sudoku" constraints in the hint. $\square$

# Appendix: Toolbox

## Week 1 Recap: Induction and Functions

**Natural numbers and induction.** We use

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

**Principle of Mathematical Induction.** Let $P(n)$ be a statement defined for all $n \in \mathbb{N}$. If

**(a)** $P(1)$ is true, and

**(b)** for all $n \in \mathbb{N}$, $P(n) \Rightarrow P(n+1)$,

then $P(n)$ is true for all $n \in \mathbb{N}$.

**Functions.** Let $A, B$ be nonempty sets. A *function* $f : A \to B$ assigns to each $a \in A$ exactly one element $f(a) \in B$. The set $A$ is the *domain* and $B$ is the *codomain*.

**Composition.** If $f : A \to B$ and $g : B \to C$ are functions, their *composition* is the function $g \circ f : A \to C$ given by

$$(g \circ f)(a) = g(f(a)) \qquad (a \in A).$$

**Image and preimage.** Let $f : A \to B$.

- For $X \subseteq A$, the *image* of $X$ under $f$ is
$$f(X) = \{f(x) : x \in X\}.$$
- For $Y \subseteq B$, the *preimage* of $Y$ under $f$ is
$$f^{-1}(Y) = \{a \in A : f(a) \in Y\}.$$

*Remark.* For a general function $f : A \to B$, the notation $f^{-1}(Y)$ denotes a preimage and does *not* require $f$ to be invertible.

**Injective / Surjective / Bijective.** Let $f : A \to B$.

- $f$ is *injective* if for all $a_1, a_2 \in A$,
$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$
- $f$ is *surjective* if $f(A) = B$, i.e. for every $b \in B$ there exists $a \in A$ such that $f(a) = b$.
- $f$ is *bijective* if it is both injective and surjective.

*Remark.* These properties depend on both the rule defining $f$ and the choice of domain $A$ and codomain $B$.

## Countability

**Finite and countable.** A set $S$ is *finite* if $S = \varnothing$ or there exists $n \in \mathbb{N}$ and a bijection $\{1, 2, \ldots, n\} \to S$. A set $S$ is *countably infinite* if there exists a bijection $\mathbb{N} \to S$. A set $S$ is *countable* if it is finite or countably infinite.

**Equivalent characterizations.** The following are standard and often useful:

- $S$ is countable $\iff$ there exists an injection $S \hookrightarrow \mathbb{N}$.
- $S$ is countable $\iff$ there exists a surjection $\mathbb{N} \twoheadrightarrow S$.

**Countability of $\mathbb{N} \times \mathbb{N}$.** There are many explicit bijections $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ (e.g. "diagonal" enumeration / pairing functions). In proofs, we often use the fact that

$$\mathbb{N} \times \mathbb{N} \text{ is countable.}$$

**Disjoint unions.** If $A$ and $B$ are countable and $A \cap B = \varnothing$, then $A \cup B$ is countable (one can interleave enumerations, or inject $A \cup B$ into $\mathbb{N} \times \{1, 2\}$). More generally, if $(A_n)_{n \in \mathbb{N}}$ are countable and pairwise disjoint, then

$$\bigcup_{n \in \mathbb{N}} A_n \text{ is countable}$$

by coding an element $x \in A_n$ as the pair $(n, k)$ where $k$ is its index in an enumeration of $A_n$.

## Relations and Equivalence Relations

**Binary relations.** A (binary) relation $\mathcal{R}$ on a set $X$ is a subset $\mathcal{R} \subseteq X \times X$. We write $x \,\mathcal{R}\, y$ instead of $(x, y) \in \mathcal{R}$.

**Equivalence relation.** A relation $\sim$ on $X$ is an *equivalence relation* if it is:

- *Reflexive:* $x \sim x$ for all $x \in X$.
- *Symmetric:* $x \sim y \Rightarrow y \sim x$.
- *Transitive:* $x \sim y$ and $y \sim z \Rightarrow x \sim z$.

**Equivalence classes.** If $\sim$ is an equivalence relation on $X$ and $x \in X$, the *equivalence class* of $x$ is

$$[x] := \{y \in X : y \sim x\}.$$

The set of all equivalence classes is denoted $X/\!\sim$.

**Partition property.** Equivalence relations and partitions are two sides of the same coin:

- The equivalence classes $\{[x] : x \in X\}$ form a partition of $X$ (classes are disjoint and their union is $X$).
- Conversely, any partition of $X$ defines an equivalence relation (declare two elements equivalent iff they lie in the same block).

**Example: congruence mod 2 on $\mathbb{Z}$.** For $a, b \in \mathbb{Z}$, define $a \sim b$ iff $b - a$ is divisible by 2. Then $\sim$ is an equivalence relation and the equivalence classes are the *even* and *odd* integers:

$$[0] = 2\mathbb{Z}, \qquad [1] = 2\mathbb{Z} + 1.$$

## Fields

**Definition (Field).** A triple $(F, +, \cdot)$ is a *field* if $F$ is a set with at least two elements and $+$ and $\cdot$ are operations on $F$ satisfying:

**Addition axioms.**

**(A1)** (Closure) If $a, b \in F$, then $a + b \in F$.

**(A2)** (Commutativity) If $a, b \in F$, then $a + b = b + a$.

**(A3)** (Associativity) If $a, b, c \in F$, then $(a + b) + c = a + (b + c)$.

**(A4)** (Identity) There exists $0 \in F$ such that $a + 0 = 0 + a = a$ for all $a \in F$.

**(A5)** (Inverses) For every $a \in F$ there exists $-a \in F$ such that $a + (-a) = (-a) + a = 0$.

**Multiplication axioms.**

**(M1)** (Closure) If $a, b \in F$, then $a \cdot b \in F$.

**(M2)** (Commutativity) If $a, b \in F$, then $a \cdot b = b \cdot a$.

**(M3)** (Associativity) If $a, b, c \in F$, then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

**(M4)** (Identity) There exists $1 \in F$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in F$.

**(M5)** (Inverses) For every $a \in F \setminus \{0\}$ there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

**Distributivity.**

**(D)** For all $a, b, c \in F$, $(a + b) \cdot c = a \cdot c + b \cdot c$.

**Basic consequences (often used without comment).** Let $(F, +, \cdot)$ be a field.

- The additive and multiplicative identities 0 and 1 are unique.
- Additive inverses and multiplicative inverses (for $a \neq 0$) are unique.
- *Additive cancellation:* if $a + b = a + c$, then $b = c$.
- *Multiplicative cancellation:* if $a \neq 0$ and $a \cdot b = a \cdot c$, then $b = c$.
- $a \cdot 0 = 0$ for all $a \in F$.
- $(-a)(-b) = ab$ and $a(-b) = -(ab)$ for all $a, b \in F$.

**Translation and dilation maps** Fix $x \in F$.

- The map $T_x : F \to F$ defined by $T_x(y) = x + y$ has inverse $T_{-x}$.
- If $x \neq 0$, the map $M_x : F \to F$ defined by $M_x(y) = x \cdot y$ has inverse $M_{x^{-1}}$.

## A Note on Finite Field Tables

In a field, $(F, +)$ is an abelian group and $(F \setminus \{0\}, \cdot)$ is an abelian group. Consequently:

- In the addition table, each row/column is a permutation of $F$.
- In the multiplication table, each *nonzero* row/column is a permutation of $F \setminus \{0\}$.

This is the "Sudoku" constraint your hint refers to.